

Task 2 : Pandas library

First: we have to get data into our environment and think carefully about what need fixing.

Here we found that age column have only 714 values out of 891

So to handle we Fill missing values in the 'Age' column with the median age

Second:

We found that embarking values was 889/891 , which misses two passengers , so to deal with this we take the most common value

Finally: the cabin values were 204/891 which is a huge data missing ,so to deal with it we dropped the column (Cabin)

Messy data

```
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   PassengerId      891 non-null    int64
1   Survived          891 non-null    int64
2   Pclass            891 non-null    int64
3   Name              891 non-null    str
4   Sex               891 non-null    str
5   Age              714 non-null    float64
6   SibSp             891 non-null    int64
7   Parch            891 non-null    int64
8   Ticket            891 non-null    str
9   Fare              891 non-null    float64
10  Cabin             204 non-null    str
11  Embarked          889 non-null    str
dtypes: float64(2), int64(5), str(5)
```

clean data

```
#   Column          Non-Null Count  Dtype
---  -
0   PassengerId      891 non-null    int64
1   Survived          891 non-null    int64
2   Pclass            891 non-null    int64
3   Name              891 non-null    str
4   Sex               891 non-null    str
5   Age              891 non-null    float64
6   SibSp             891 non-null    int64
7   Parch            891 non-null    int64
8   Ticket            891 non-null    str
9   Fare              891 non-null    float64
10  Embarked          891 non-null    str
dtypes: float64(2), int64(5), str(5)
```

3.

Analysis & Insights

After cleaning the data, a clear picture emerged. Survival on the Titanic was not purely a matter of luck; it was heavily influenced by specific demographic and socioeconomic factors:

A. The Impact of Gender: Gender was the absolute strongest predictor of survival, heavily reflecting the strict "women and children first" evacuation protocol:

Female Survival Rate: 74.20% .

Male Survival Rate: 18.89%. .

```
Total no of women surv: 233
Survival Percentage by Gender:
Sex
female      74.203822
male        18.890815
Name: Survived, dtype: float64
```

B. The Impact of Social Class (Ticket Class): Socioeconomic status played a critical role in who made it to the lifeboats. Passengers in higher classes had better access to the upper decks and lifeboats:

1st Class Survival Rate: ~63% .

3rd Class Survival Rate: ~24% .

Insight: Wealthier passengers in first class had a significantly higher chance of surviving compared to the working-class passengers in third class.

```
Survival Rate by Passenger Class:  
Pclass  
1      62.962963  
2      47.282609  
3      24.236253  
Name: Survived, dtype: float64
```

C. The Impact of Age: By grouping the passengers into different life stages (Children, Teenagers, Adults, and Seniors), we found that age also dictated evacuation priority:

Children (0-12 years): Had the highest survival rate at **~58%**.

Seniors (60+ years): Had the lowest survival rate at **~23%**.

Insight: Children were prioritized during the rescue, while the elderly were the least likely to survive.

Survival Percentage by Age Group:

AgeGroup

Child (0-12) 57.971014

Teenager (13-18) 42.857143

Adult (19-60) 36.575342

Senior (60+) 22.727273